

The Wet Stack / Mireye

Physical-world intelligence for data-center siting and EPC change control. Technical document, Round 1.

What We're Building

The Wet Stack is a construction and EPC intelligence layer built on **Mireye**, covering the two moments where a data center gets decided: once at site selection, and again every time equipment is substituted mid-build. Both decisions run on physical facts. Our job is to make every one of those facts traceable back to the Mireye record it came from, rather than quietly backfilled.

AI agents plan investigations and write explanations; a deterministic Python layer performs every calculation, unit conversion, gate check and score. A regression test enforces this: there is no path by which the model can emit a number.

How We Use Mireye

Mireye is the ground truth for the entire product. Of **3,819 evidence records** currently in the system, **3,435 come from the live Mireye API**, drawn through it from USGS, FEMA, NREL, EIA, NOAA, EPA, NRCS and FCC.

- **Catalog-driven, not guessed** - we read `/v1/meta/fields` and bind our 30 concepts to the live catalog's **310 fields** by name. 11 map exactly, 6 need a unit conversion applied at the boundary, 3 are labelled contextual proxies, and 2 sit in the billed `parcel_record` group and are opt-in.
- **We corrected the contract against the live service** - our first integration got five things wrong: catalog entries key on name not key; geocode returns `lat/lng/accuracy_type`; `/v1/fetch` takes coordinates or an address but never both; absence arrives as `value: null` inside `fields{}`, not a separate list; and confidence is a word, not a float. All five are written up in `docs/mireye-contract.md`.
- **We respect what it costs** - Mireye bills per field per location and `parcel_record` is 300 credits, so responses are cached against the catalog's own `ttl_seconds` and a budget caps spend per investigation. Hitting the cap is not an error: unqueried sites record a gap saying so, rather than presenting partial coverage as complete.
- **Endpoints in use** - `/v1/meta/fields`, `/v1/geocode` and `/v1/fetch` for anything that feeds a score; `/v1/ask` for exploratory questions only. A test fails if the scoring path ever calls `/v1/ask`, because prose must never become a number in a calculation.
- **Where Mireye has no equivalent, we say so** - 14 of our 30 concepts have no catalog field. Six are filled from public datasets; the rest stay open with a named next action, and are filed back as feature requests.

Architecture & Core Stack

- **Backend** - FastAPI + Pydantic v2 for the API and schema layer, Pint for unit-safe physical and electrical arithmetic, LangGraph for multi-step agent orchestration, PyMuPDF for PDF ingestion and OCR, Redis for caching Mireye responses, Neo4j for impact-graph traversal, Postgres/pgvector for optional live-mode vector storage.
- **Frontend** - React 18 + TypeScript on Vite, Tailwind for the design system, Recharts for engineering margin visualisation, Leaflet for GIS site mapping.
- **Four agents** - Change Orchestrator (substitution lifecycle), Site Intelligence Supervisor (LangGraph state machine across 8 dimensions), Project Knowledge / Document MCP agent (submittal search with page-level citations), and a Requirement Parser that turns plain-language equipment asks into structured physics and electrical constraints.
- **Both workflows run the same seven-step loop** - Understand, Plan, Evidence, Signals, Replan, Impact, Action - spanning **30 tracked concepts across 8 dimensions**: terrain, water, power, connectivity, civil, hazards, environmental and regulatory.

Evidence Integrity

The system is built on one rule: what was not measured stays unmeasured. It is never rounded to zero and never quietly replaced by a nearby measurement. Three mechanisms enforce it.

- **Relevance validation** - a value sharing a unit with the one required but not its meaning is stored as `CONTEXTUAL_PROXY`: cited and visible, but unable to fill the field, close its gap or pass a gate. Mireye's 0.4% design **wet-bulb** against our **dry-bulb** requirement is the reference case, both in Celsius. Enforcing it dropped coverage from **0.348 to 0.259** and left the decision unchanged. That drop is the system being honest.
- **Gaps become actions** - a concept with no source is flagged **MISSING** with the action that would resolve it, such as commissioning a geotechnical survey. Eight such concepts are deliberately kept because each produces an action; six others were removed because they produced noise without one.
- **OCR discipline** - values read by OCR are recorded at half confidence and flagged for manual verification, because a misread digit is a fabricated figure with a citation attached.

What's Actually Built and Live

Change Intelligence (during construction)

- **9 deterministic verification gates** - manufacturer comparability, voltage and phase match, refrigerant suitability, rated ambient against the site's ASHRAE extremes taken from Mireye, footprint boundary, structural floor loading and others.
- **15 Pint-checked deltas** across weight, maximum support point load, length, width, height, footprint area, voltage, phases, FLA, MCA, MOCP, power input, refrigerant type, refrigerant charge and cooling capacity.
- Design margins, cost and schedule impact, cascade loading across concurrent changes, catalog ranking across **11 equipment classes**, a dependency impact graph, and an RFI and action-package generator with citation export.
- **Every case resolves deterministically to one of three states**: NEEDS INFORMATION, ENGINEER REVIEW, or first-pass checks closed.

Site Intelligence (before construction)

- Deterministic, explainable scoring across all 8 dimensions, every score carrying its Mireye citation and an explicit list of open data gaps. On the demo project: 9 candidate sites, 200 open gaps of which 9 are blocking.

Live system numbers

- **3,435 live Mireye evidence records**, and **97.8% of all 3,819 records are real sourced data** rather than synthetic.
- **5 contract mismatches** found and fixed against the live Mireye service.
- **277 tests passing**; lint and typecheck clean.
- **5 public datasets wired in** alongside Mireye, covering 6 concepts the catalog does not serve: PeeringDB, EPA/USGS Water Quality Portal, USGS PAD-US, EIA-861 and the FEMA National Risk Index.

Impact - What the System Has Already Caught

- **Harbour Point, VA** - groundwater at 6,610 mg/L total dissolved solids, which is brackish and directly changes the cooling design. It would have been invisible behind a placeholder value.
- **Rio Verde Mesa, AZ** - rated Very High for wildfire and sitting inside Tonto National Forest, confirmed by two independent datasets. It also corrected our own earlier error, which had reported the site as 4 km from a shooting range because a capped query never returned the national forest it sits in.
- **FEMA scale correction** - FEMA's wildfire index runs 0 to 100, but its own Very Low band has a median of 43.6, nowhere near zero. Scored linearly, every safe site in the country would have lost half its wildfire points. Thresholds now follow FEMA's published bands, with a test that fails if a future release moves them.

Why it matters: a hyperscale site is a \$1-3 billion commitment made on the weakest evidence in the project's lifecycle. The cost of choosing badly is not the land; it is the brackish water or wildfire exposure found after the option is signed. During construction, this identifies which discipline needs review while explicitly declining to claim adequacy. That refusal is what makes it usable by someone carrying real liability.

Plan for the Next Days

- Replace the remaining 84 synthetic document records with AHRI-certified ratings and manufacturer submittals, through the upload and OCR path already built.
- Bring in 4 more public sources: WRI Aqueduct, USGS 3DEP, USDA CropScape, NOAA ISD.
- File the concepts Mireye does not yet serve back as feature requests, so the catalog and the model converge rather than diverge.
- Move to a production architecture: Kafka, Redis, Postgres and a projection-based impact graph, designed in docs/production-11d.md.

Implementation Detail

Decision precedence (change_orchestrator.py) - every case walks the same three checks, in order, before it can close:

1. A required parameter is missing or unevidenced, giving **NEEDS INFORMATION** and opening an information gap.
2. Otherwise a verification gate fails, a delta threshold is breached, or a site design limit taken from Mireye is exceeded, giving **ENGINEER REVIEW** and triggering impact-graph traversal.
3. Otherwise first-pass checks are closed, within all design limits.

Catalog engine (catalog_engine.py) weights the 11 equipment classes: capacity 25%, energy efficiency 20%, electrical compatibility 15%, climate fit 10%, footprint 10%, then water efficiency, cost, maintenance and project compatibility at 5% each. A criterion with an unknown input is not scored and the total renormalises, so a product with no published capacity cannot out-rank one that simply scores badly.

Cascade engine (cascade.py) aggregates concurrent changes on electrical load, structural weight and cooling duty against substation, generator and roof-steel headroom. Headroom is a fraction of a capacity, so where the facility's capacity is not stated it returns **NEEDS_INFORMATION** and names what is missing, rather than issuing a structural verdict against an assumed building.

Core module map - gates.py (9 gates), deltas.py / units.py (15 deltas), margins.py, impact.py, cost_schedule.py, and adapters/mireye.py (typed contract, caching, credit budget). Agents: change_orchestrator.py, workflow.py / planner.py, knowledge_agent.py.

Core Dependency Versions

Layer	Library	Version	Role
Backend	FastAPI	>=0.110	Async API layer, OpenAPI contract generation
Backend	Pydantic v2	>=2.6	Schema validation and environment config
Backend	Pint	>=0.23	Unit-safe physical and electrical arithmetic
Backend	LangGraph	>=0.2	Stateful multi-agent orchestration
Backend	PyMuPDF	>=1.24	PDF ingestion, table extraction, OCR
Backend	Redis	>=5.0	Mireye response cache, keyed on the catalog's own TTLs
Backend	Neo4j / Psycopg 3	>=5.19 / >=3.1	Impact graph traversal / Postgres + pgvector
Frontend	React 18 + Vite 6	-	Typed UI, Tailwind v4 styling, dev and build tooling